

## FRELIGHSBURG, QC, Canada

WMO: 713730

Lat: 45.0394N

Lon: 72.8561W

Elev: 735

StdP: 14.31

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -11.6      | -6.2 | -20.3                         | 1.8 | -10.4 | -15.0 | 2.5 | -5.1 | 18.8                  | 24.7 | 17.2 | 23.6 | 4.6                   | 270  | 0.484 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 16.9                   | 84.8            | 71.1 | 82.1 | 69.7 | 79.6 | 68.3 | 73.9                | 81.3 | 72.1 | 78.9 | 70.5 | 77.0 | 6.0                  | 190  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 71.6                            | 119.8 | 77.9 | 69.8 | 112.8 | 76.0 | 68.2 | 106.5 | 74.7 | 38.0          | 81.6 | 36.3 | 79.1 | 34.9 | 77.0 | 81.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 16.0              | 13.8 | 12.0 | -19.2                      | 89.5 | 6.7                | 2.3 | -24.0                                              | 91.1 | -27.9      | 92.4 | -31.7      | 93.7 | -36.6      | 95.3 |    |    |
|                   |      |      | -19.5                      | 77.5 | 6.5                | 2.1 | -24.1                                              | 79.1 | -27.9      | 80.3 | -31.5      | 81.5 | -36.2      | 83.1 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 44.7   | 17.2  | 19.9  | 29.0  | 42.8  | 55.8  | 64.3  | 68.9  | 67.3  | 60.3  | 48.2  | 36.5  | 24.7  |
|                                                                               | DBStd       | 20.46  | 13.89 | 12.05 | 11.44 | 9.26  | 8.10  | 6.78  | 5.26  | 5.50  | 7.56  | 8.88  | 10.18 | 11.32 |
|                                                                               | HDD50       | 4158   | 1016  | 842   | 655   | 254   | 34    | 1     | 0     | 0     | 8     | 145   | 417   | 786   |
|                                                                               | HDD65       | 7789   | 1481  | 1262  | 1115  | 668   | 302   | 93    | 22    | 40    | 179   | 523   | 854   | 1250  |
|                                                                               | CDD50       | 2226   | 1     | 0     | 5     | 38    | 214   | 430   | 586   | 535   | 316   | 89    | 12    | 0     |
|                                                                               | CDD65       | 385    | 0     | 0     | 0     | 2     | 18    | 73    | 142   | 110   | 38    | 2     | 0     | 0     |
|                                                                               | CDH74       | 2457   | 0     | 0     | 1     | 24    | 149   | 531   | 917   | 625   | 204   | 6     | 0     | 0     |
|                                                                               | CDH80       | 476    | 0     | 0     | 0     | 3     | 24    | 120   | 189   | 107   | 33    | 0     | 0     | 0     |
| Wind                                                                          | WSAvg       | 5.3    | 5.8   | 5.8   | 6.1   | 6.2   | 5.3   | 4.6   | 4.3   | 4.0   | 4.4   | 5.1   | 5.8   | 5.6   |
| Precipitation                                                                 | PrecAvg     | 45.2   | 3.0   | 2.6   | 3.0   | 3.2   | 4.0   | 4.3   | 4.7   | 4.4   | 4.2   | 4.2   | 3.5   | 4.0   |
|                                                                               | PrecMax     | 53.5   | 6.8   | 7.3   | 5.0   | 6.5   | 7.8   | 7.6   | 6.6   | 8.3   | 7.7   | 7.2   | 4.9   | 7.9   |
|                                                                               | PrecMin     | 35.9   | 1.3   | 1.3   | 1.9   | 0.7   | 1.8   | 1.4   | 3.1   | 1.9   | 1.3   | 1.7   | 2.1   | 2.0   |
|                                                                               | PrecStd     | 4.6    | 1.2   | 1.3   | 0.7   | 1.5   | 1.6   | 1.6   | 1.0   | 1.3   | 1.8   | 1.7   | 0.8   | 1.3   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 56.4  | 52.5  | 66.2  | 77.7  | 83.2  | 88.0  | 88.3  | 86.7  | 84.7  | 74.5  | 65.4  | 53.0  |
|                                                                               |             | MCWB   | 52.5  | 45.3  | 56.2  | 60.6  | 66.0  | 73.3  | 74.9  | 73.0  | 72.1  | 62.2  | 57.3  | 49.0  |
|                                                                               | 2%          | DB     | 47.1  | 45.6  | 55.0  | 69.7  | 78.3  | 83.6  | 84.6  | 82.9  | 79.1  | 70.1  | 60.6  | 48.3  |
|                                                                               |             | MCWB   | 43.4  | 40.3  | 47.2  | 55.8  | 62.5  | 69.7  | 71.9  | 70.6  | 68.9  | 60.7  | 53.6  | 44.7  |
|                                                                               | 5%          | DB     | 40.8  | 41.0  | 49.4  | 63.3  | 73.9  | 80.0  | 81.9  | 80.2  | 75.8  | 65.8  | 56.2  | 44.0  |
|                                                                               |             | MCWB   | 37.9  | 36.8  | 42.4  | 50.9  | 60.6  | 67.8  | 70.2  | 69.1  | 66.8  | 57.9  | 50.5  | 41.1  |
|                                                                               | 10%         | DB     | 35.9  | 36.9  | 44.9  | 58.0  | 69.9  | 76.7  | 79.1  | 77.5  | 72.3  | 61.7  | 51.7  | 39.3  |
|                                                                               |             | MCWB   | 33.1  | 33.3  | 39.3  | 47.8  | 58.6  | 66.0  | 68.8  | 68.0  | 64.9  | 55.4  | 46.7  | 36.3  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 53.0  | 46.8  | 57.6  | 63.0  | 69.5  | 75.5  | 77.3  | 75.3  | 73.4  | 65.9  | 59.4  | 50.1  |
|                                                                               |             | MCDB   | 56.5  | 50.0  | 67.2  | 74.3  | 78.3  | 84.4  | 85.9  | 83.2  | 81.2  | 70.4  | 63.2  | 52.6  |
|                                                                               | 2%          | WB     | 44.1  | 40.9  | 48.2  | 57.6  | 66.0  | 72.5  | 74.3  | 72.8  | 70.6  | 62.0  | 54.9  | 45.4  |
|                                                                               |             | MCDB   | 46.0  | 44.9  | 53.7  | 67.0  | 73.5  | 80.3  | 81.3  | 79.7  | 76.9  | 68.2  | 59.3  | 47.5  |
|                                                                               | 5%          | WB     | 37.8  | 36.7  | 43.5  | 52.9  | 63.2  | 70.2  | 72.3  | 71.1  | 68.3  | 58.8  | 50.8  | 41.2  |
|                                                                               |             | MCDB   | 40.2  | 40.4  | 47.9  | 60.9  | 70.5  | 77.1  | 78.8  | 77.4  | 73.8  | 64.4  | 55.0  | 43.9  |
|                                                                               | 10%         | WB     | 33.3  | 33.6  | 39.7  | 48.9  | 60.2  | 67.7  | 70.7  | 69.5  | 66.1  | 56.0  | 46.7  | 36.7  |
|                                                                               |             | MCDB   | 35.5  | 36.6  | 44.8  | 56.5  | 67.5  | 73.8  | 77.0  | 75.4  | 71.7  | 60.8  | 50.9  | 39.1  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 15.9  | 16.3  | 15.6  | 17.2  | 18.4  | 17.4  | 16.9  | 16.9  | 17.0  | 14.8  | 13.2  | 13.3  |
|                                                                               |             | MCDBR  | 18.6  | 18.0  | 21.2  | 25.7  | 23.2  | 21.8  | 19.6  | 19.2  | 19.4  | 19.3  | 19.5  | 15.8  |
|                                                                               |             | MCWBR  | 17.1  | 14.8  | 14.9  | 15.4  | 12.8  | 11.7  | 10.1  | 10.1  | 11.1  | 12.4  | 14.8  | 14.6  |
|                                                                               | 5% WB       | MCDBR  | 18.0  | 17.4  | 18.9  | 22.6  | 19.6  | 18.2  | 16.5  | 16.7  | 17.2  | 17.5  | 17.9  | 15.9  |
|                                                                               |             | MCWBR  | 17.0  | 15.3  | 14.8  | 15.3  | 12.6  | 11.0  | 9.8   | 9.8   | 10.9  | 12.5  | 15.5  | 15.3  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.262  | 0.284 | 0.319 | 0.349 | 0.370 | 0.418 | 0.427 | 0.405 | 0.365 | 0.327 | 0.303 | 0.281 |       |
|                                                                               | taud        | 2.381  | 2.305 | 2.357 | 2.392 | 2.389 | 2.251 | 2.234 | 2.291 | 2.396 | 2.510 | 2.497 | 2.381 |       |
|                                                                               | Ebn at Noon | 274    | 286   | 290   | 289   | 285   | 270   | 266   | 268   | 271   | 269   | 257   | 252   |       |
|                                                                               | Edh at Noon | 23     | 30    | 34    | 36    | 37    | 43    | 43    | 39    | 32    | 25    | 21    | 21    |       |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 453    | 740   | 1113  | 1365  | 1602  | 1752  | 1773  | 1582  | 1263  | 732   | 450   | 322   |       |
|                                                                               | RadStd      | 52     | 77    | 83    | 164   | 162   | 143   | 133   | 75    | 101   | 71    | 58    | 28    |       |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)  | (e)                     | (f)   | (g)     | (h)    | (i)   | (j)     | (k)   | (l) | (m)         |       |       |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (46 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | +112  | N/A   |

Nomenclature: See separate page